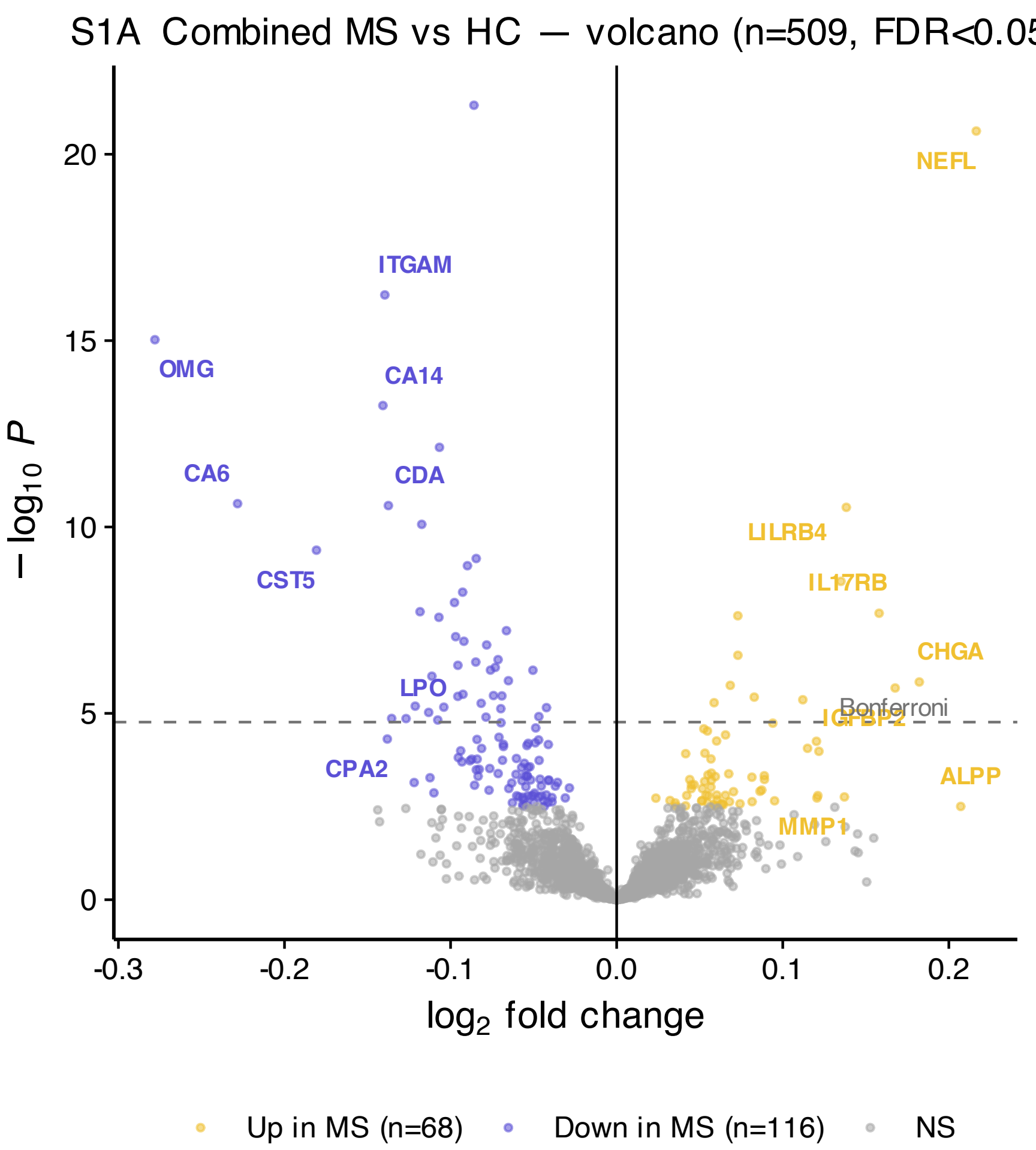
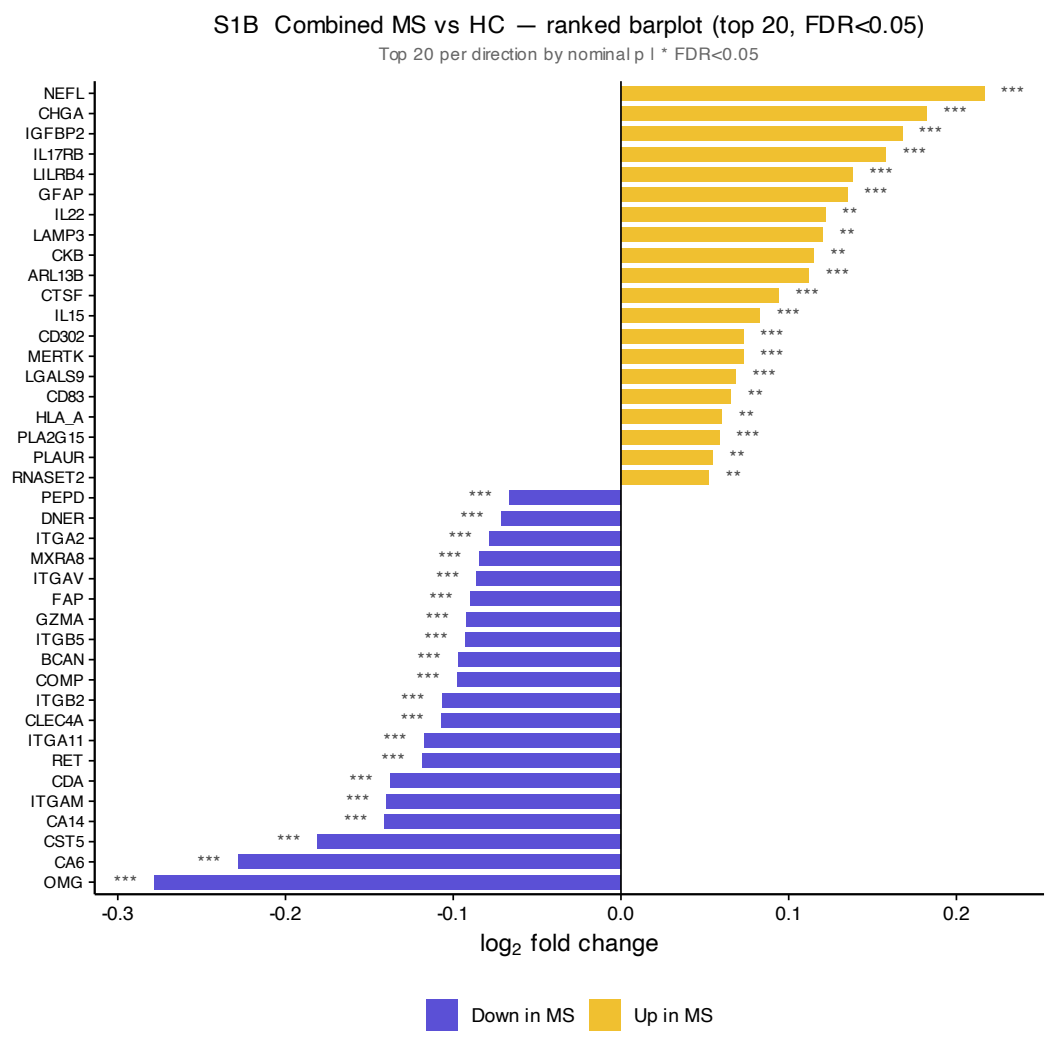
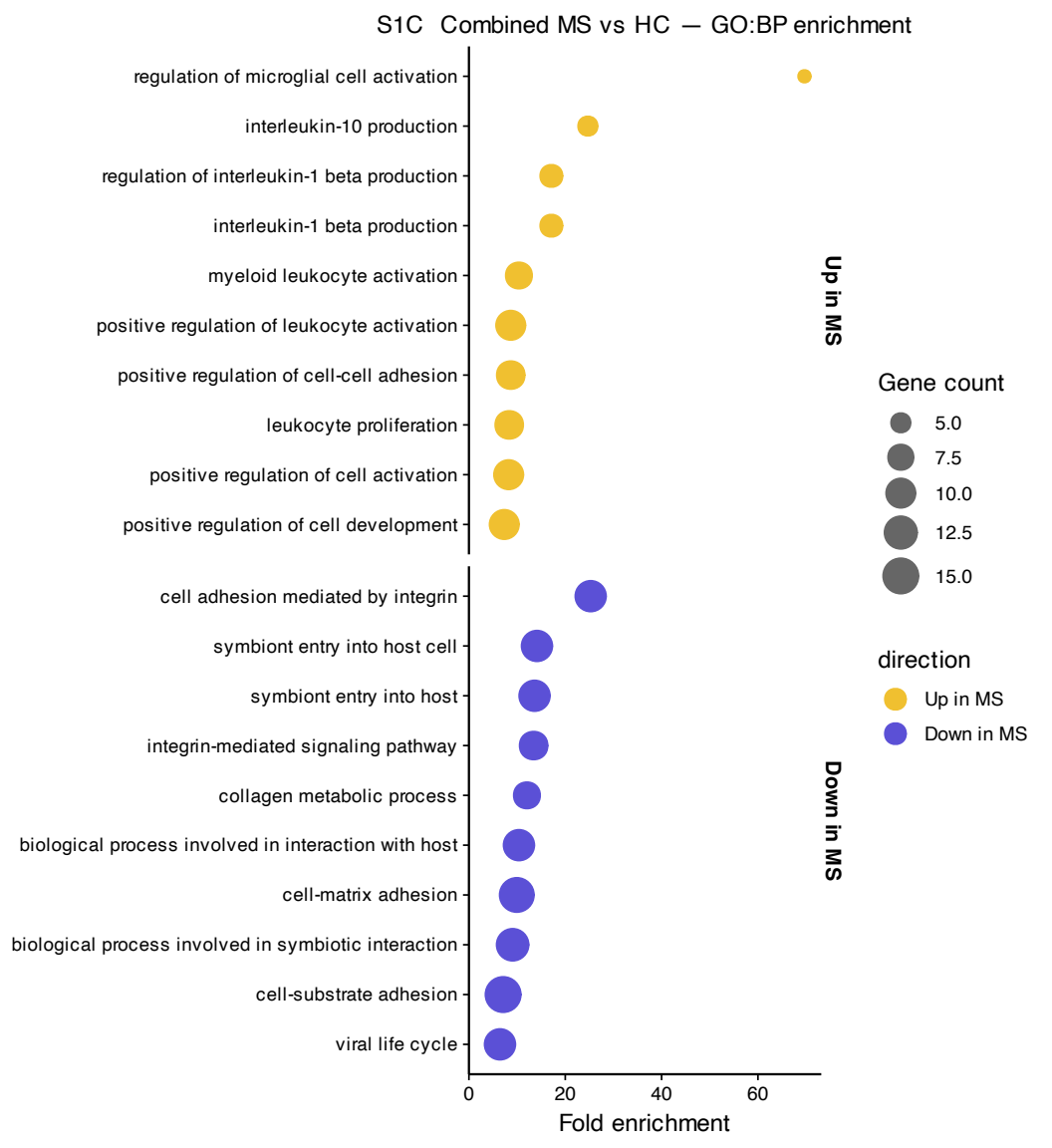
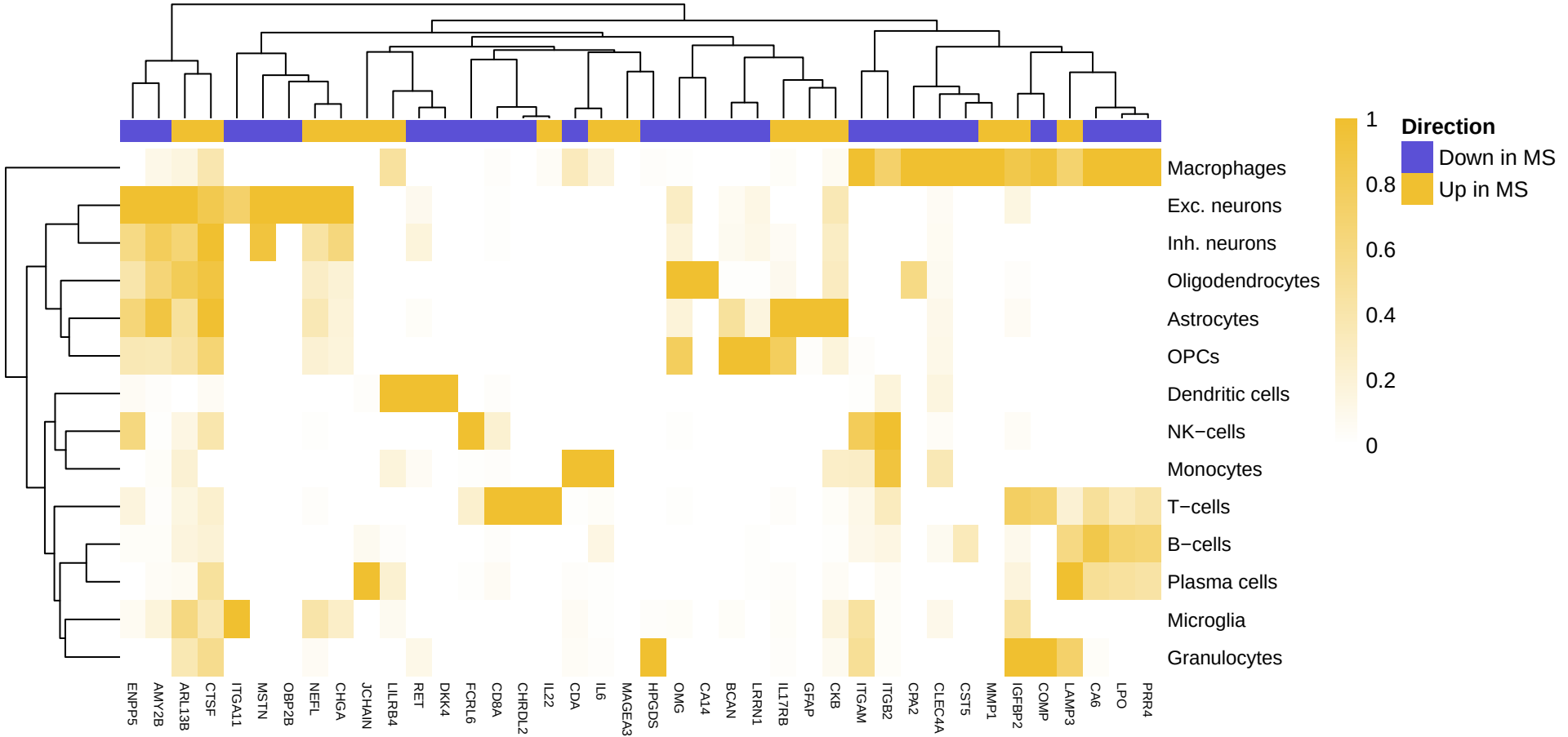
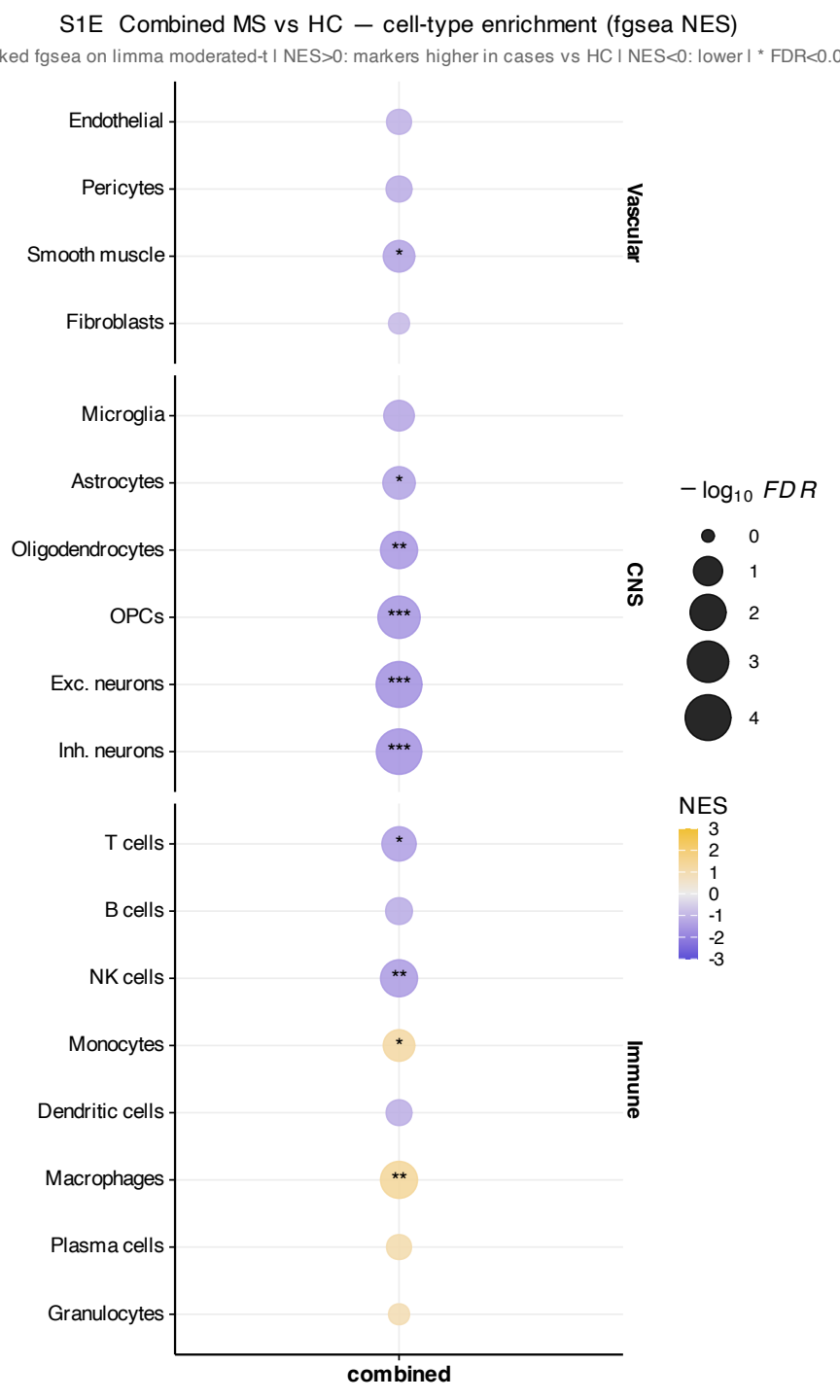

S1A Combined MS vs HC — volcano (n=509, FDR<0.05
NEFL
ITGAM
OMG
CA14
CA6
CDA
LILRB4
CST5
IL17RB
CHGA
LPO
Bonferroni
IGFBP2
CPA2
ALPP
MMP1
20
15
10
5
0
-0.3
-0.2
-0.1
0.0
0.1
0.2
$-\log_{10} P$
$\log_2$ fold change
Up in MS (n=68)
Down in MS (n=116)
NS

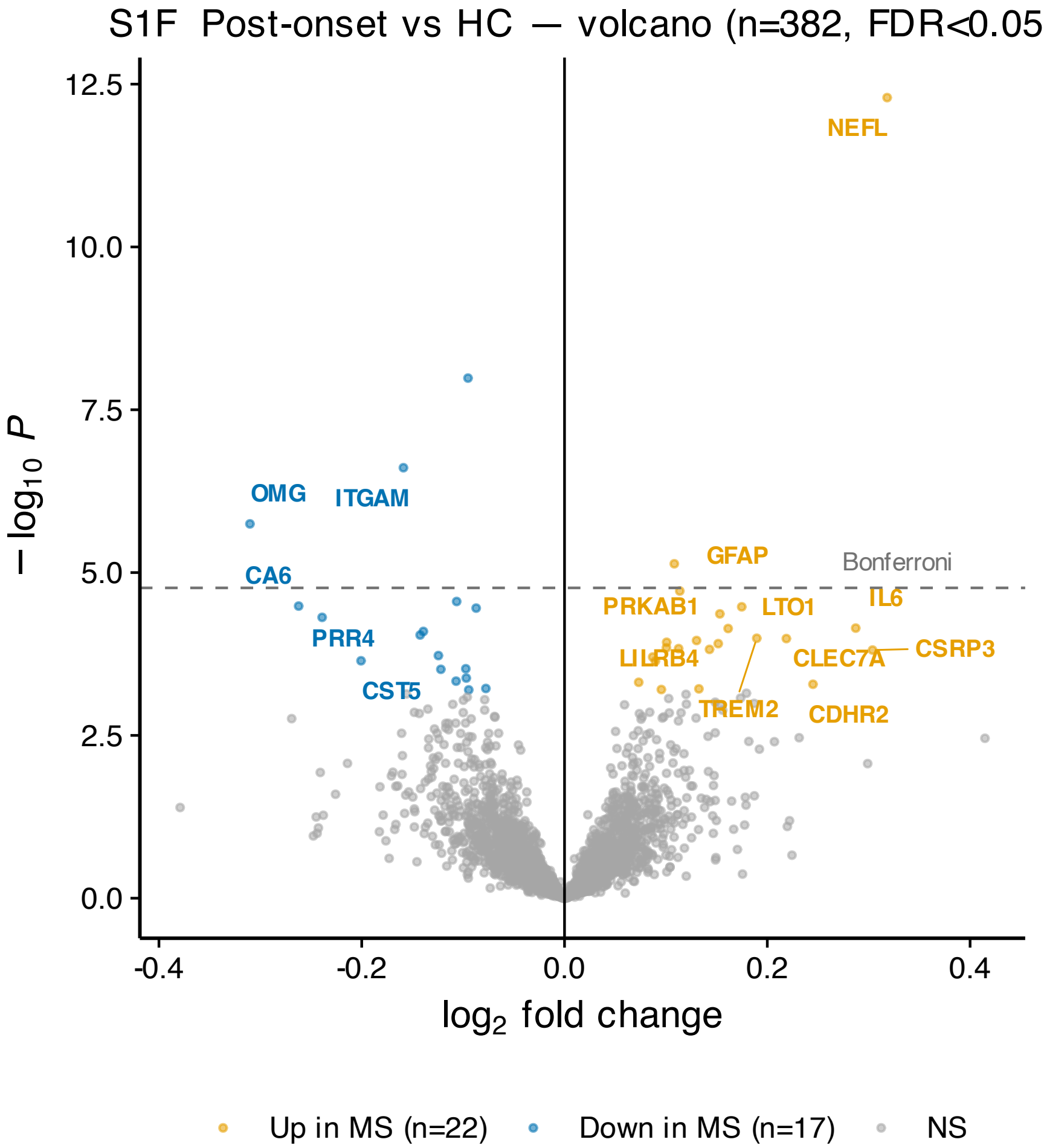

S1B Combined MS vs HC — ranked barplot (top 20, FDR<0.05)
Top 20 per direction by nominal p | * FDR<0.05
NEFL
CHGA
IGFBP2
IL17RB
LILRB4
GFAP
IL22
LAMP3
CKB
ARL13B
CTSF
IL15
CD302
MERTK
LGALS9
CD83
HLA_A
PLA2G15
PLAUR
RNASET2
PEPD
DNER
ITGA2
MXRA8
ITGAV
FAP
GZMA
ITGB5
BCAN
COMP
ITGB2
CLEC4A
ITGA11
RET
CDA
ITGAM
CA14
CST5
CA6
OMG
-0.3
-0.2
-0.1
0.0
0.1
0.2
log2 fold change
Down in MS
Up in MS

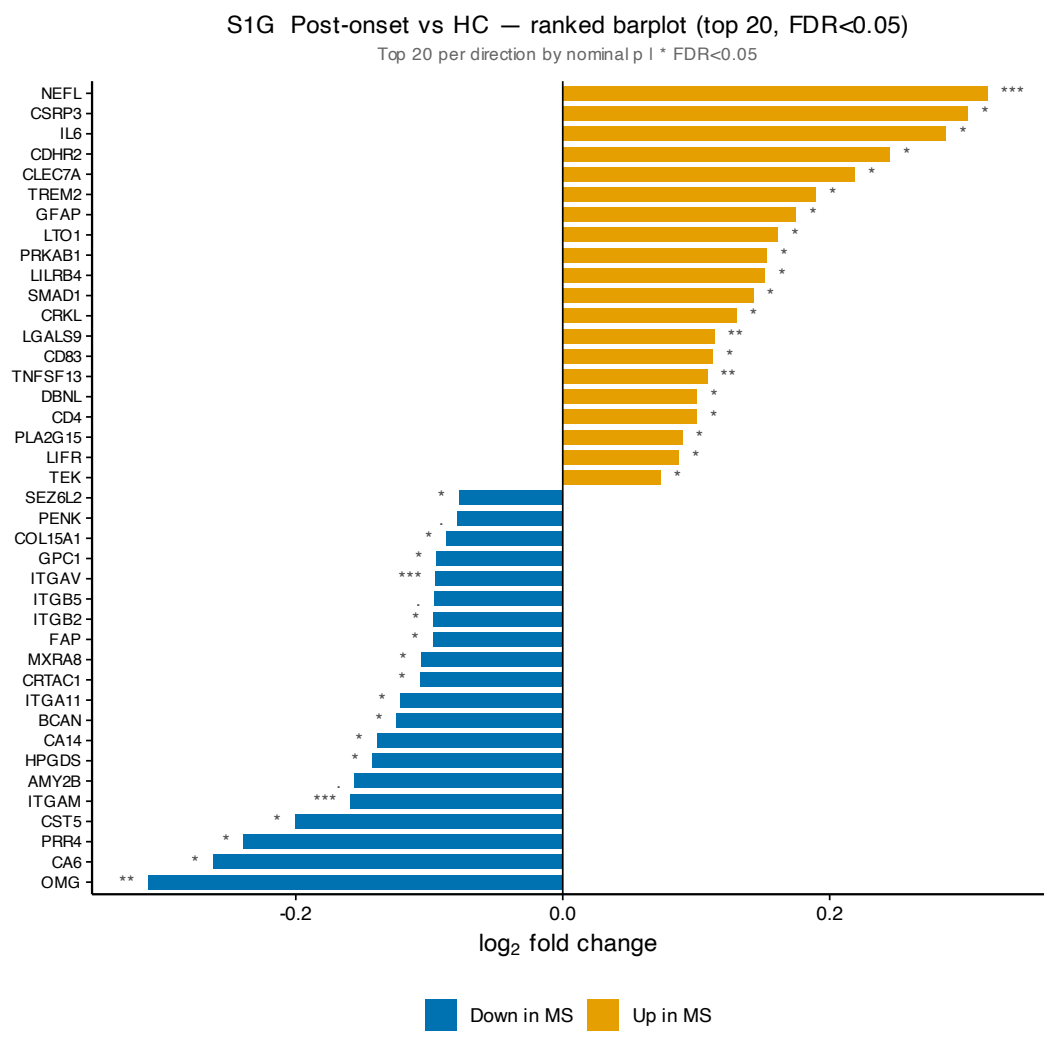

S1C Combined MS vs HC — GO:BP enrichment
regulation of microglial cell activation
interleukin-10 production
regulation of interleukin-1 beta production
interleukin-1 beta production
myeloid leukocyte activation
positive regulation of leukocyte activation
positive regulation of cell-cell adhesion
leukocyte proliferation
positive regulation of cell activation
positive regulation of cell development
cell adhesion mediated by integrin
symbiont entry into host cell
symbiont entry into host
integrin-mediated signaling pathway
collagen metabolic process
biological process involved in interaction with host
cell-matrix adhesion
biological process involved in symbiotic interaction
cell-substrate adhesion
viral life cycle
Up in MS
Down in MS
0
20
40
60
Fold enrichment
Gene count
5.0
7.5
10.0
12.5
15.0
direction
Up in MS
Down in MS

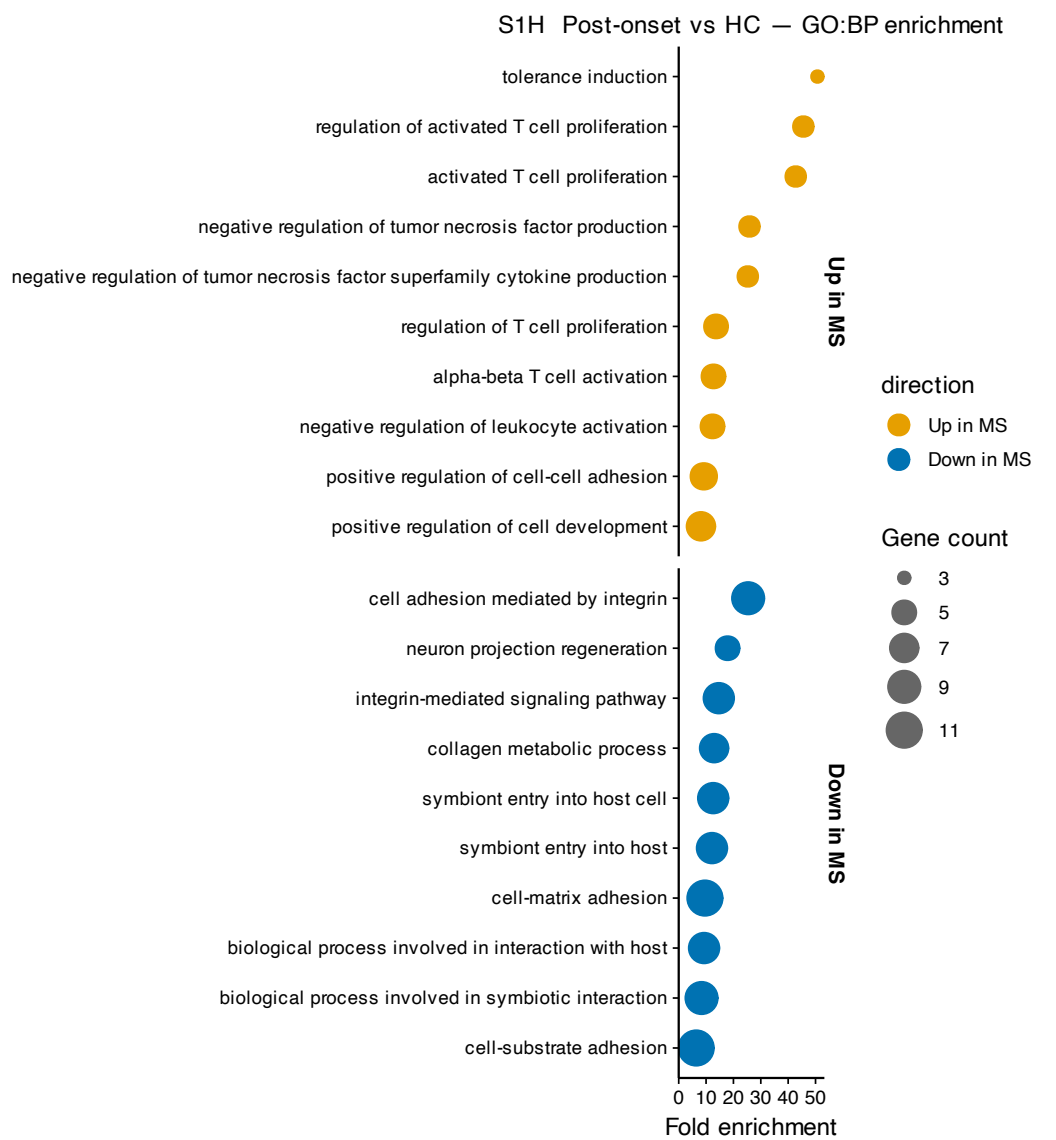
S1D Combined MS vs HC − HPA cell−type expression (top 40 DEPs)



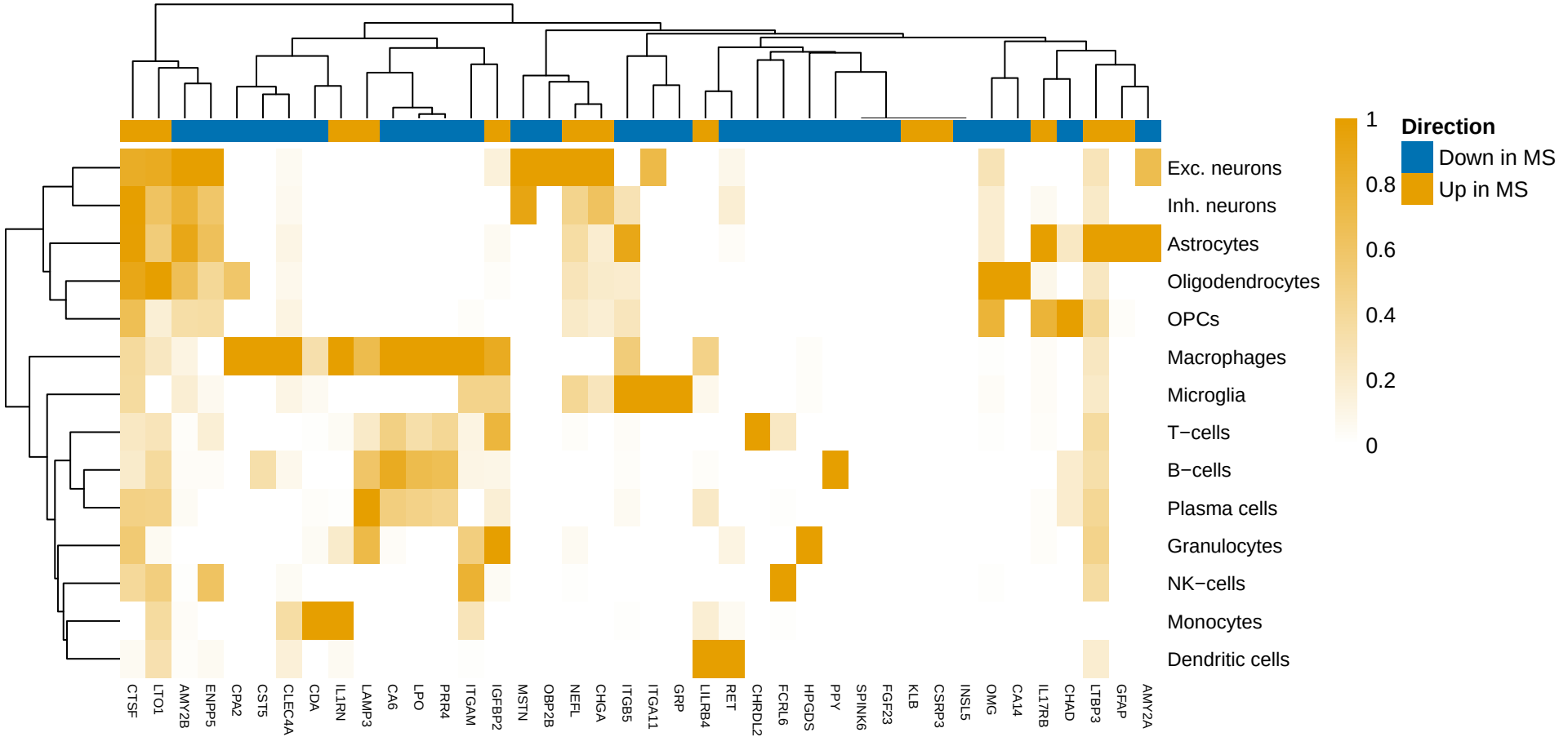

S1E Combined MS vs HC — cell-type enrichment (fgsea NES)
ked fgsea on limma moderated-t | NES>0: markers higher in cases vs HC | NES<0: lower | * FDR<0.0
Endothelial
Pericytes
Smooth muscle
Fibroblasts
Microglia
Astrocytes
Oligodendrocytes
OPCs
Exc. neurons
Inh. neurons
T cells
B cells
NK cells
Monocytes
Dendritic cells
Macrophages
Plasma cells
Granulocytes
Vascular
CNS
Immune
combined
− log10 FDR
0
1
2
3
4
NES
3
2
1
0
-1
-2
-3

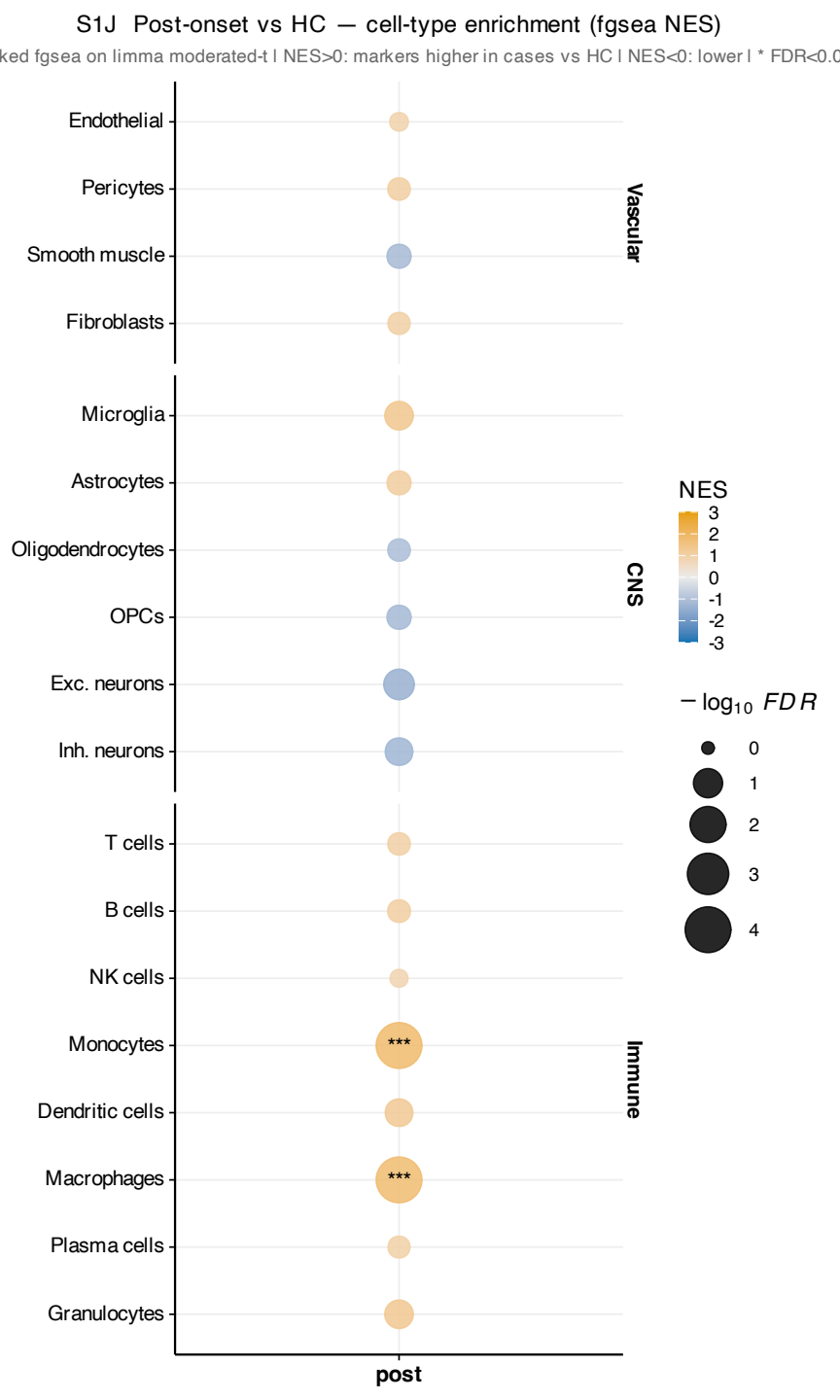

S1F Post-onset vs HC — volcano (n=382, FDR<0.05
12.5
10.0
7.5
5.0
2.5
0.0
−log10 P
-0.4
-0.2
0.0
0.2
0.4
log2 fold change
NEFL
OMG
ITGAM
CA6
PRR4
CST5
GFAP
Bonferroni
PRKAB1
LTO1
IL6
LILRB4
CLEC7A
CSRP3
TREM2
CDHR2
Up in MS (n=22)
Down in MS (n=17)
NS

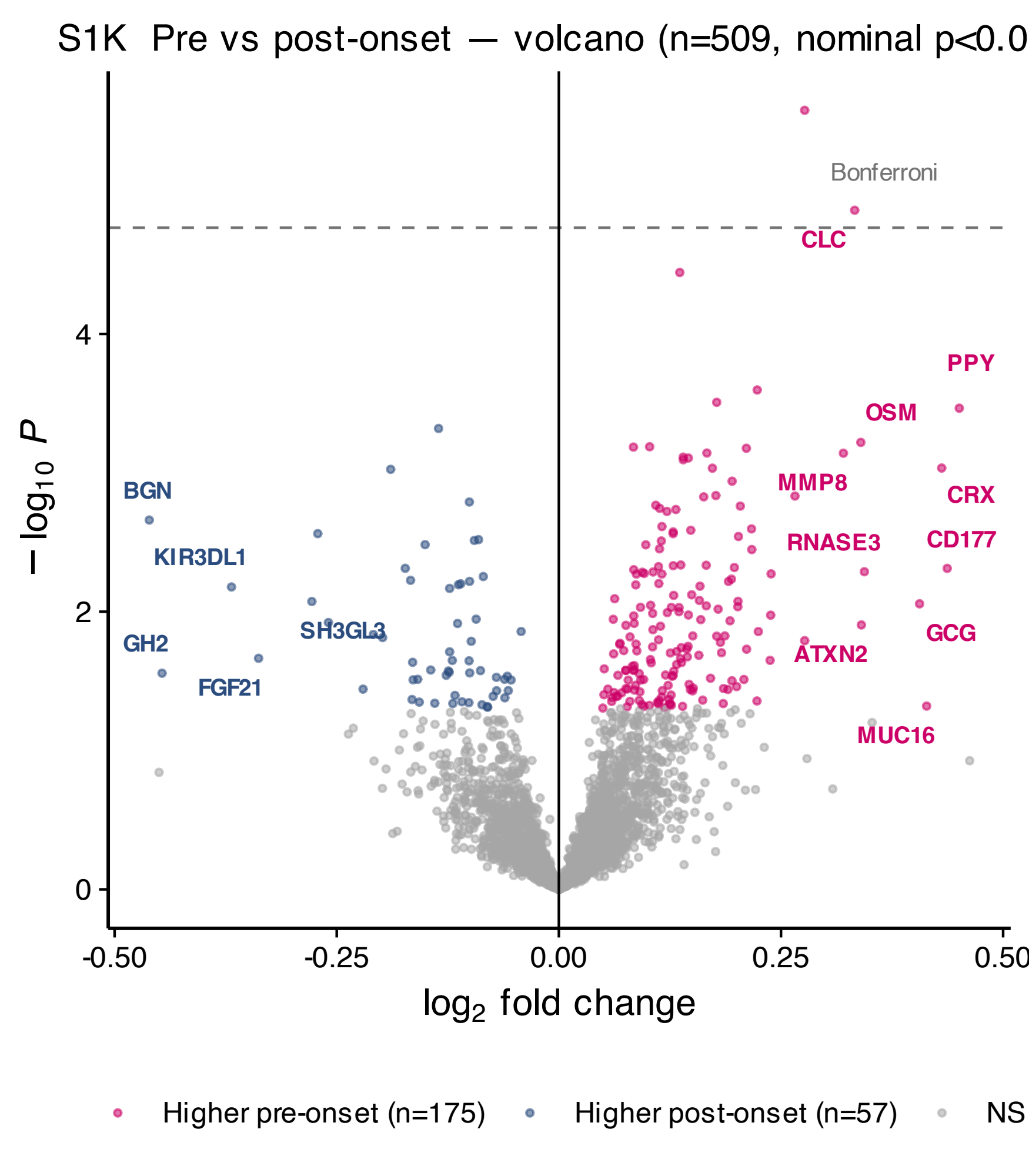

S1G Post-onset vs HC — ranked barplot (top 20, FDR<0.05)
Top 20 per direction by nominal p | * FDR<0.05
NEFL
CSRP3
IL6
CDHR2
CLEC7A
TREM2
GFAP
LTO1
PRKAB1
LILRB4
SMAD1
CRKL
LGALS9
CD83
TNFSF13
DBNL
CD4
PLA2G15
LIFR
TEK
SEZ6L2
PENK
COL15A1
GPC1
ITGAV
ITGB5
ITGB2
FAP
MXRA8
CRTAC1
ITGA11
BCAN
CA14
HPGDS
AMY2B
ITGAM
CST5
PRR4
CA6
OMG
-0.2
0.0
0.2
log2 fold change
Down in MS
Up in MS

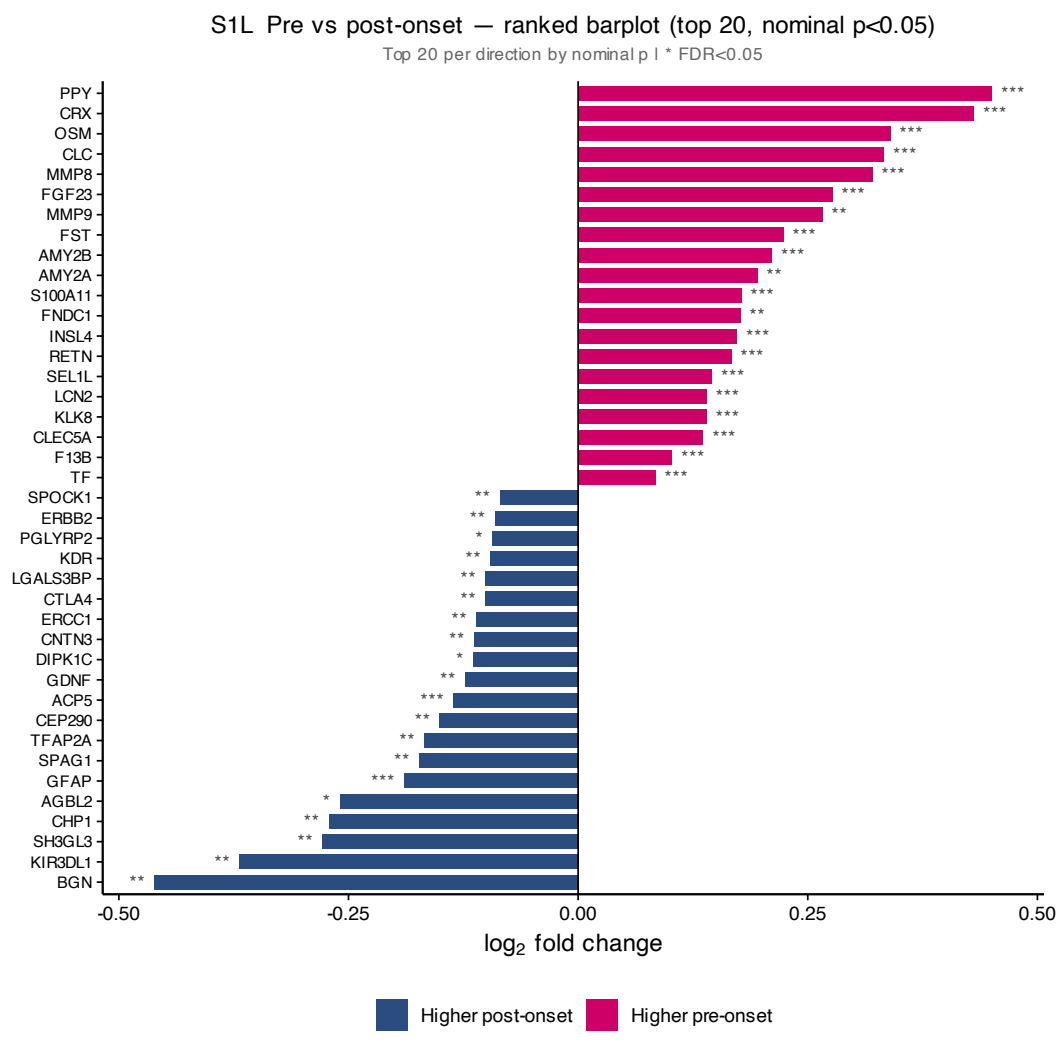

S1H Post-onset vs HC — GO:BP enrichment
tolerance induction
regulation of activated T cell proliferation
activated T cell proliferation
negative regulation of tumor necrosis factor production
negative regulation of tumor necrosis factor superfamily cytokine production
regulation of T cell proliferation
alpha-beta T cell activation
negative regulation of leukocyte activation
positive regulation of cell-cell adhesion
positive regulation of cell development
Up in MS
cell adhesion mediated by integrin
neuron projection regeneration
integrin-mediated signaling pathway
collagen metabolic process
symbiont entry into host cell
symbiont entry into host
cell-matrix adhesion
biological process involved in interaction with host
biological process involved in symbiotic interaction
cell-substrate adhesion
Down in MS
0 10 20 30 40 50
Fold enrichment
direction
Up in MS
Down in MS
Gene count
3
5
7
9
11

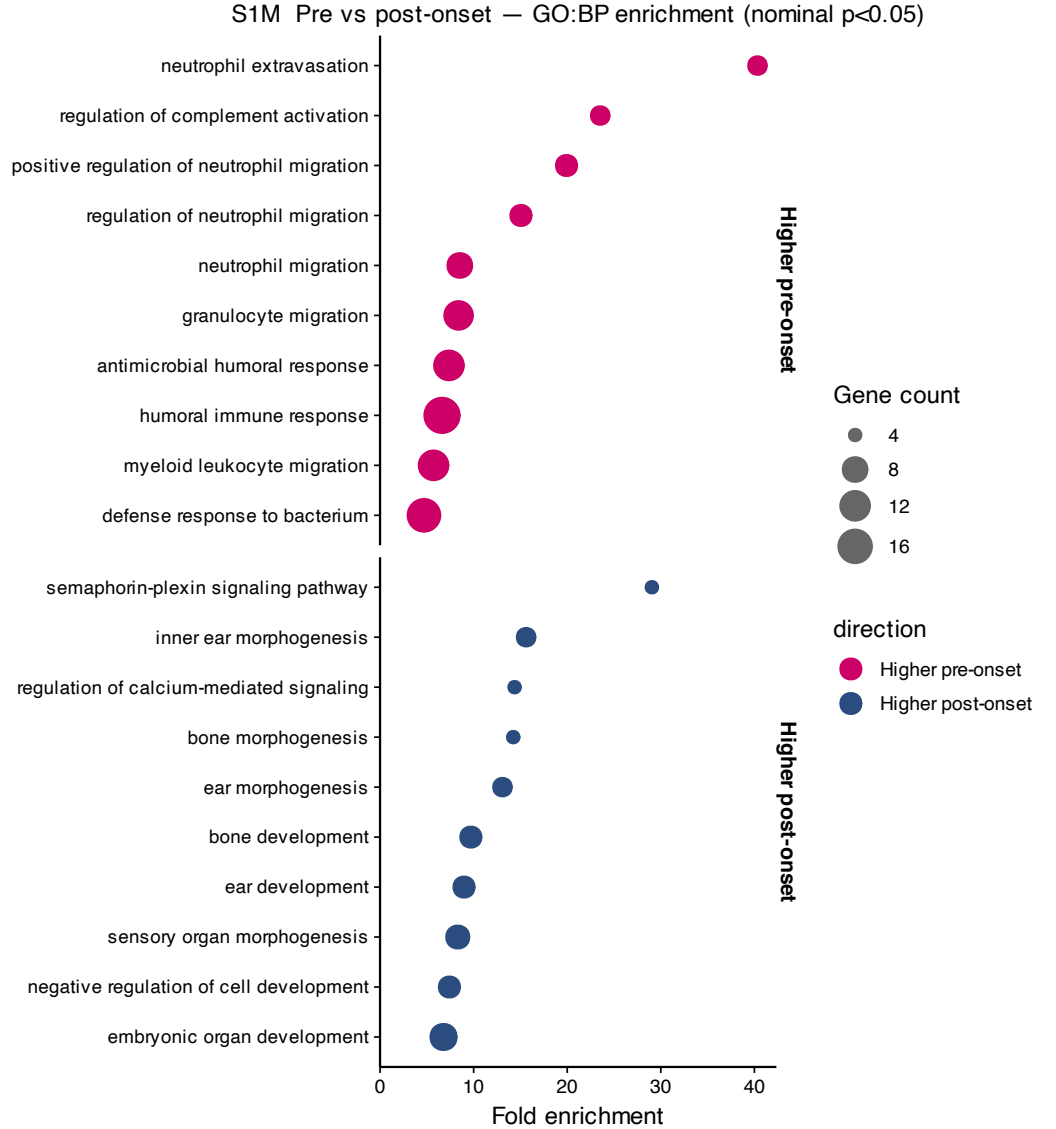
S1I Post−onset vs HC − HPA cell−type expression (top 40 DEPs)

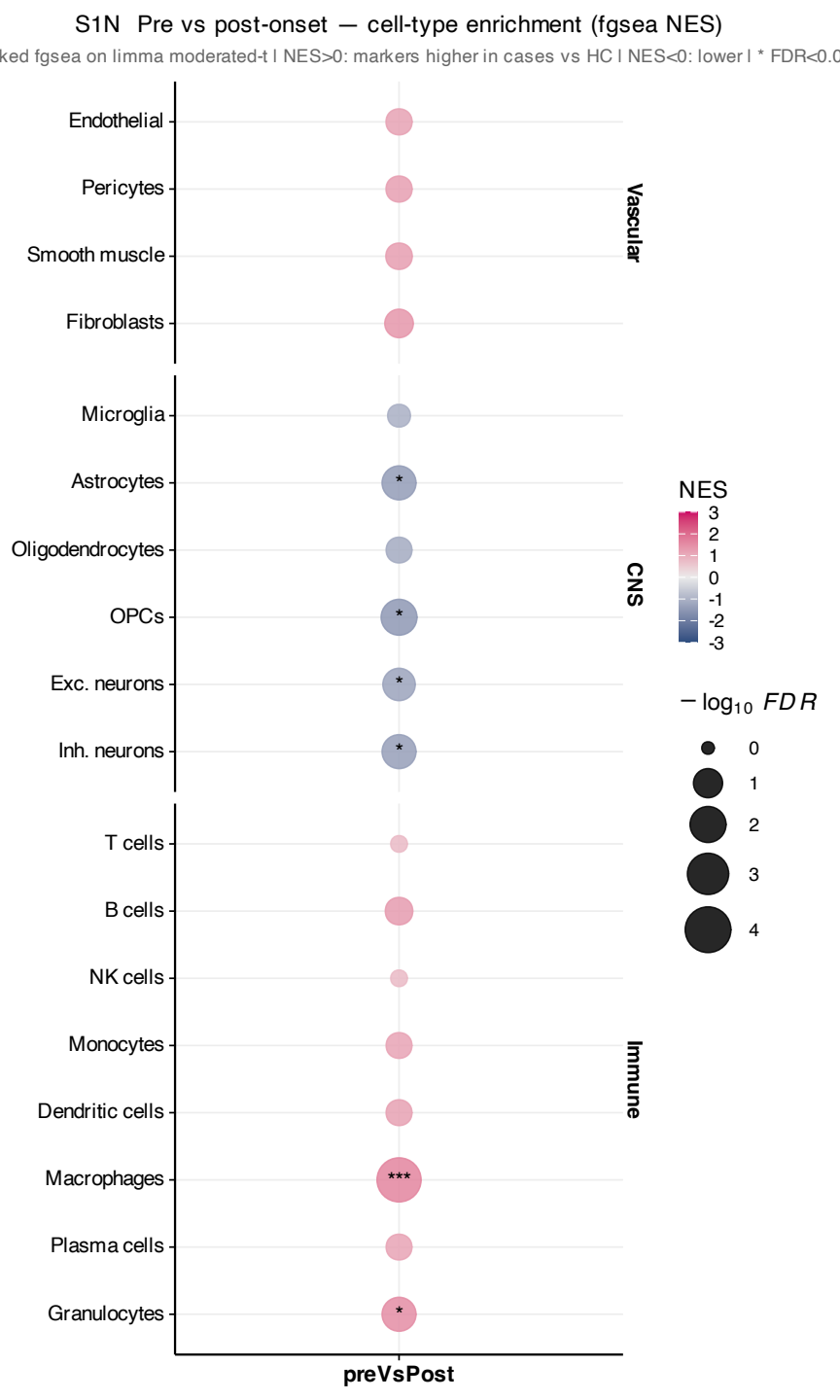



S1J Post-onset vs HC — cell-type enrichment (fgsea NES)
ked fgsea on limma moderated-t | NES>0: markers higher in cases vs HC | NES<0: lower | * FDR<0.0
Endothelial
Pericytes
Smooth muscle
Fibroblasts
Vascular
Microglia
Astrocytes
Oligodendrocytes
OPCs
Exc. neurons
Inh. neurons
CNS
T cells
B cells
NK cells
Monocytes
Dendritic cells
Macrophages
Plasma cells
Granulocytes
Immune
***
***
post
NES
3
2
1
0
-1
-2
-3
− log10 FDR
0
1
2
3
4


S1K Pre vs post-onset — volcano (n=509, nominal p<0.0
Bonferroni
CLC
PPY
OSM
MMP8
CRX
RNASE3
CD177
GCG
ATXN2
MUC16
BGN
KIR3DL1
SH3GL3
GH2
FGF21
4
2
0
-0.50
-0.25
0.00
0.25
0.50
$-\log_{10} P$
$\log_2$ fold change
Higher pre-onset (n=175)
Higher post-onset (n=57)
NS


S1L Pre vs post-onset — ranked barplot (top 20, nominal p<0.05)
Top 20 per direction by nominal p | * FDR<0.05
PPY
CRX
OSM
CLC
MMP8
FGF23
MMP9
FST
AMY2B
AMY2A
S100A11
FNDC1
INSL4
RETN
SEL1L
LCN2
KLK8
CLEC5A
F13B
TF
SPOCK1
ERBB2
PGLYRP2
KDR
LGALS3BP
CTLA4
ERCC1
CNTN3
DIPK1C
GDNF
ACP5
CEP290
TFAP2A
SPAG1
GFAP
AGBL2
CHP1
SH3GL3
KIR3DL1
BGN
-0.50
-0.25
0.00
0.25
0.50
log2 fold change
Higher post-onset
Higher pre-onset


S1M Pre vs post-onset — GO:BP enrichment (nominal p<0.05)
neutrophil extravasation
regulation of complement activation
positive regulation of neutrophil migration
regulation of neutrophil migration
neutrophil migration
granulocyte migration
antimicrobial humoral response
humoral immune response
myeloid leukocyte migration
defense response to bacterium
Higher pre-onset
semaphorin-plexin signaling pathway
inner ear morphogenesis
regulation of calcium-mediated signaling
bone morphogenesis
ear morphogenesis
bone development
ear development
sensory organ morphogenesis
negative regulation of cell development
embryonic organ development
Higher post-onset
0
10
20
30
40
Fold enrichment
Gene count
4
8
12
16
direction
Higher pre-onset
Higher post-onset


S1N Pre vs post-onset — cell-type enrichment (fgsea NES)
ked fgsea on limma moderated-t | NES>0: markers higher in cases vs HC | NES<0: lower | * FDR<0.0
Endothelial
Pericytes
Smooth muscle
Fibroblasts
Vascular
Microglia
Astrocytes
Oligodendrocytes
OPCs
Exc. neurons
Inh. neurons
CNS
T cells
B cells
NK cells
Monocytes
Dendritic cells
Macrophages
Plasma cells
Granulocytes
Immune
preVsPost
NES
3
2
1
0
-1
-2
-3
− log10 FDR
0
1
2
3
4